TeachPrint

3rd Grade Math: Multiplication — Expert

Advanced application with multi-step reasoning, unknowns, and challenge problems.

Name: _______________________________ Date: _______________

A. Multi-Step Multiplication

1. A school orders 7 boxes of pencils. Each box has 24 pencils. The school gives 38 pencils to classrooms. How many pencils remain? ______

2. A theater has 8 rows with 36 seats in each row. If 47 seats are reserved for guests, how many seats remain? ______

3. A bakery makes 9 trays with 28 muffins on each tray. It sells 125 muffins. How many muffins are left? ______

B. Unknowns & Reasoning

1. A number multiplied by 7 equals 84. What is the number? ______

2. Which number makes the equation true? 6 × ____ + 5 = 41. ______

3. A rectangle has 6 rows of 14 tiles. How many tiles are there? If 18 tiles break, how many good tiles remain? ______

C. Challenge Problems

1. A teacher has 6 packs of 18 stickers. She gives 4 stickers to each of 17 students. How many stickers are left? Show your work.

2. A school has 5 buses. Each bus carries 42 students. On a trip, 17 students are absent. The remaining students are split equally among 7 activity groups. How many students are in each group? ______

3. Create a problem that requires multiplying first and then subtracting. Your final answer must be 100 or less. Write the problem and solve it.

D. Explain Your Thinking

1. Explain why 8 × 7 and 7 × 8 have the same product. Use words, an array, or a drawing.

TeachPrint — Answer Key

A. Multi-Step Multiplication

1. 130 pencils
2. 241 seats
3. 127 muffins

B. Unknowns & Reasoning

1. 12
2. 6
3. 66 good tiles

C. Challenge Problems

1. 40 stickers
2. 193 students; 193 ÷ 7 is not a whole number, so the situation does not produce an equal whole-number split. Students should identify this issue.
3. Answers vary; check that multiplication happens before subtraction and the final answer is 100 or less.

D. Explain Your Thinking

Answers vary. Switching the number of rows and items per row changes the arrangement but not the total, illustrating the commutative property.